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The 2013 mass public school closure and firearm violence in Chicago: A quasi-experimental difference-in-differences analysis

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ABSTRACT

School closure continues to be a contentious policy decision in cities around the United States, with opponents emphasizing negative effects of closure on urban neighborhoods. Little research has been done on the association of school closure with firearm violence, a major challenge affecting US cities. We sought to determine the impact of the 2013 Chicago mass public school closure on firearm violence in neighborhoods affected by building vacancy following closure. We used an inverse probability of treatment weighted design to balance treatment and control groups. The study period was from 2010 to 2019. The unit of analysis was an individual school, with the primary exposure being immediate building vacancy. The primary outcome was shootings with secondary outcomes of weapons violations and assaults/battery with a firearm, drawn from administrative data from the Chicago Police Department. School closure was associated with a 9.88% increase in shootings near the school location (95% CI: 3.59-16.99; p = 0.001). Long term building vacancy was associated with a 10.20% increase in shootings near the building location (95% CI, 2.73-18.84, p = 0.005). Secondary outcomes of assaults/battery with a firearm and weapons violations also demonstrated significant increases. These results demonstrate that school building vacancy following closure is associated with an increase in shootings in the surrounding neighborhood. In decisions around school closure, policymakers should consider the multifactorial impacts of closure on communities and potential community-driven productive re-use of space, particularly given closure frequently occurs in already-disinvested neighborhoods.

1. Introduction

In 2013, Chicago Public Schools shuttered 49 elementary schools, primarily in the city's South and West Sides (Tieken and Auldridge-Reveles, 2019; Ewing, 2018). This event was the largest mass public school closure in U.S. history, though mass closures occurred around the same time in other U.S. cities like Philadelphia and Detroit (Ewing, 2018; Brazil and Candipan, 2022; Royal, 2022; Royal and Cothorne, 2021). Across cities, the stated goal of school districts was to reduce budget deficits by closing under-enrolled schools (Ewing, 2018). The closures in Chicago generated 45 vacant elementary school buildings (and one additional vacant non-school building) in the immediate aftermath of the closures. The school district and city aimed to resell and redevelop vacant properties; but a 2023 analysis by the Chicago Sun-Times demonstrated that 10 years later, 26 buildings affected by the closures remained vacant (Fitzpatrick et al., 2023). Research has demonstrated that these school closures primarily affected poor students of color, and particularly Black students (Ewing, 2018; Grant et al., 2017). Moreover, studies documented increased inequality of access to education as well as decreased short-term educational outcomes for displaced students (Lee and Lubienski, 2017; Pearman and Marie Greene, 2022). Beyond the loss of educational access, an important aspect of school closure that has received less attention is the potential impact of closure on the safety of neighborhoods and communities surrounding the former schools (Good, 2022; Tieken and Auldridge-Reveles, 2019).

In Chicago, the school closures have been highly contentious – both historically and presently – due to additional community concerns about increased crime and violence in surrounding neighborhoods (Good, 2022; Brazil, 2020). Similar to school closures, firearm violence is spatially concentrated in disadvantaged, racially minoritized neighborhoods (Polcari et al., 2023; Uzzi et al., 2023) . This is of particular importance in Chicago, where firearm violence is a primary driver of the life expectancy gap between White and Black residents (Bishop-Royse et al., 2024). Nationally, firearm violence is now the leading cause of death amongst U.S. children (Goldstick et al., 2022; Villarreal et al.,

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2024), which has resulted in schools being a key vehicle for many community violence intervention (CVI) strategies. Many community members thus argued that school closures could increase violence both through increased neighborhood blight, loss of valuable educational resources, and loss of community organizing hubs for CVI, among other mechanisms.

Prior research on the relationship between school closure and community violence has been limited in scope. Brazil (2020) notably examined the 2013 Chicago mass closure by comparing census tract crime rates for schools that closed (n = 49) to schools that were on the list for potential closure but remained open (n = 79). He found small short-term reductions in both violent and nonviolent crime lasting less than 1 and 2 years, respectively (Brazil, 2020). Studies in Philadelphia and Scotland also demonstrated small, short-term reductions in crime following closure (Steinberg et al., 2019; Borbely et al., 2023). One possible mechanism for this is grounded in routine activities theory, which posits that increased crime will occur at the intersection of likely offenders, suitable targets, and the absence of capable guardians or social control (Miró, 2014). Schools have previously been identified as locations that generate all three of these necessary components, particularly when students are unsupervised by teachers or school administrators such as after school (Roman, 2002). Thus, closing a school might be expected to decrease crime in the surrounding area. However, it is unclear whether this corresponds with the more specific category of firearm violence, which is distinct from other crimes like property damage, vandalism, and theft.

Other studies, however, have found increased crime following closure (Brinig and Garnett, 2009). Brinig and colleagues examined the closure of 130 Catholic elementary schools in Chicago between 1984 and 2004, finding that closures were associated with higher crime and lower social cohesion. As such, researchers have theorized that schools may function to increase neighborhood social cohesion—the degree of social connection, trust, and cooperation among residents—which has been recognized as a key protective factor against community violence (Rich and Owens, 2023, Sampson et al., 1997). It is possible that school closures could erode neighborhood social cohesion over time, as social bonds weaken without a shared community hub. Brazil also theorized that crime "may emerge or become stronger the longer a building remains vacant," consistent with theories that neighborhood blight can increase risk of crime (Brazil, 2020). For example, substantial research has documented that greening a vacant lot has been associated with reductions in firearm violence near the lot (Branas et al., 2011; Moyer et al., 2019; South et al., 2023). No studies to date have examined the impacts of school closure on community firearm violence beyond the short-term window, particularly in the context of persistent vacancy 10 years after school buildings were vacated.

While existing theoretical mechanisms provide rationale for the link between school closures and crime overall, there are several that emphasize the relevance of firearm violence in particular. First, firearm violence is important and distinct from other forms of crime in that it is lethal, and can precipitate cyclical disadvantage in a neighborhood (Bancalari et al., 2022; Kravitz-Wirtz et al., 2022). For example, the loss of life has direct and profound impacts on a neighborhood's social environment, fracturing trust, reducing social cohesion, and limiting social capital (Smith, 2015; Kravitz-Wirtz et al., 2022). Similarly, firearm violence has been linked to higher community-level trauma and fear within the community (Gaylord-Harden et al., 2022; Sharpe et al., 2025). These responses can be associated with adverse coping behaviors (e.g., firearm carriage, isolation from neighbors) that subsequently perpetuate and amplify firearm violence (Motley et al., 2017; Richardson et al., 2020;Rich and Grey, 2005; Lee et al., 2025).

In this Chicago-based study, we examine the relationship between the building vacancies generated by the 2013 public school closures and firearm violence. Utilizing a inverse probability of treatment weighted spatial difference-in-differences analysis (2010-2019), we examined associations between school building vacancy and annual changes in rates of shootings, weapons violations, and aggravated assaults/batteries with firearms. We hypothesized a relative increase in firearm violence over the 10-year study period in neighborhoods surrounding closed schools compared with neighborhoods unaffected by school closure. We further hypothesized that prolonged vacancy, defined as building vacancy 10 years following closure would be associated with larger effects.

2. Methods

2.1. Data and variables

We used administrative data from Chicago Public Schools (CPS) and incident-level crime data from the Chicago Police Department (CPD) to conduct a inverse probability of treatment weighted difference-in-differences (DiD) analysis evaluating associations between the 2013 mass school closures and firearm violence-related outcomes: shootings, weapons violations, and assaults/batteries with a firearm (Chicago Open Data Portal, 2025; South et al., 2023). The study period (2010-2019) was selected due to the availability of comprehensive shooting data beginning in 2010. The period ends in 2019 before the onset of the COVID-19 pandemic to avoid bias from non-parallel shocks. COVID-19 likely had highly heterogeneous impacts across neighborhoods on firearm violence and on school operations, including differing transitions to remote learning (Sun et al., 2022). Because these effects could violate the parallel trends assumption, post-COVID years were excluded. Sensitivity analyses were conducted on time periods from 2010 to 2017 and from 2010 to 2023 to test sensitivity of results to different time periods.

Publicly available data from the CPD identified all shooting incidents in Chicago (n = 59,637), weapons violations (n = 57,470), and assault/battery incidents with a firearm (n = 57,676) between January 2010 and December 2019, including the date, time, address, and geocoded location (Chicago Open Data Portal, 2025). We generated point-based and area-based geographic metrics of these incidents. We used incident geocoordinates to calculate annual kernel density estimates (KDEs) of incidents per square mile, applying a 0.5 mile bandwidth to capture neighborhood-level effects (Branas et al., 2011; Moyer et al., 2019). Kernel density estimation is a widely-utilized method that allows for the smoothing of individual incidents, by weighting incidents occurring closer to a given location more highly than those further from that location (Moyer et al., 2019; Zewdie et al., 2025). The spatstat package in R was used for KDE estimation (Baddeley et al., 2015). In sensitivity analyses, the bandwidth of the kernel density estimate was varied from 0.25 miles to 2 miles (Appendix Table 1).

2.2. Sample identification and balancing

In the primary analysis, elementary school buildings that became vacant as a result of the 2013 closures (n = 45) were the treatment group, while public elementary schools that remained open (n = 405) served as potential controls. This treatment definition was chosen in order to eliminate the inclusion of schools that were publicized as closing but actually experienced merger with another school, meaning that the school building itself remained open. Only public, non-alternative schools were eligible for inclusion, as alternative school types like charter or magnet schools were largely unaffected by closure. This allows us to isolate the effect of complete loss of public school activity and subsequent building vacancy on firearm violence and minimize treatment heterogeneity. Because school closures disproportionately occurred in low-income Black neighborhoods, we implemented an energy-based inverse probability of treatment weighting procedure to ensure the parallel trends assumption was fulfilled (Huling and Mak, 2024). Energy balanced weights were used because they remain well-behaved under the square-root transformation required for spatial panel estimation, preserving comparability in the outcome scale between treated and control groups and supporting interpretable effect estimates. Specifically, we weighted schools on characteristics of their associated census tracts: the social vulnerability index, the percentage of the population that was Black, the percentage of the population that was Hispanic, the poverty rate, the percent of individuals 19 and younger, and the percent of parents in the labor force (Center for Disease Control and Prevention, 2013). All variables were based on 2013 American Community Survey statistics, representing the end of the pre-treatment period.

In a second and third model, we classified treatment to specify if buildings affected by initial vacancy remained vacant through 2023 (n = 26) or received re-use following closure (n = 19). This analysis was conducted to better determine the association of both potential states, repurposed and vacant, with firearm violence. Due to the potential for systematic differences between the neighborhoods of vacant buildings that remained vacant or received repurposing, reweighting of control schools against the selected treated schools was performed. Schools that received the alternative treatment state were removed from the control group for the weighting procedure and analysis in this model to avoid contamination.

2.3. Model specification and statistical analysis

We applied a two-way fixed-effects spatial panel regression model to determine the relationship between the 2013 mass school closures and selected measures of firearm violence. Spatial regression models were used to correct for spatial error and spatial lag in the data given the proximity of schools in the dataset (Marca et al., 2026). This supports the underlying Stable Unit Treatment Values Assumption in difference-in-difference models. School-level fixed effects were used to further control time-invariant differences between schools (Steinberg et al., 2019). The year 2014 was used as the first year of treatment as closures took effect in the fall of 2013 and thus this was the first full year affected by closure. The spatial structure of the school locations was represented through a k-nearest-neighbor weights matrix including observations of the k = 5 nearest neighbors, employing functions from the spdep package (Roger, 2022).

Our outcome variable, Y_{it}, was the KDE of shootings associated with each school's surrounding area. The parallel trends assumption was tested first by plotting the trends in KDE across both groups during the pre-intervention period. Then, following a procedure described by South et al. (2023) we evaluated the interaction terms for the following regression model to determine if the interaction term, β_2, differed significantly from zero:

$$Y_{it} = \lambda WY_{it} + \beta_1 t + \beta_2 Closure_i \times t + \delta_i + \mu_t + \varepsilon_{it} \qquad \text{Eq. 1}$$

The term $Closure_i$ is an indicator for whether school i was vacant and t represents time in years. Fixed effects were included at the school (δ_i) in order to capture time-invariant school-level differences that may bias the estimate. μ_t represents year fixed effects. λ represents the spatial coefficient capturing the degree of spatial dependence amongst units and W is a spatial weights matrix of known constants, this term captures spatial lag in the model. ε_{it} is a vector of spatially autocorrelated residuals. Models were further adjusted for spatial error. A non-zero value of β_3 indicates that treated and control schools exhibited different trends during the pre-intervention period. Although the parallel trends assumption cannot be directly tested, finding no significant pre-intervention difference in trends provides evidence consistent with the assumption's plausibility.

The effect of school closure was assessed using the following regression model:

$$Y_{it} = \lambda WY_{it} + \beta_1 PrePost_t + \beta_3 Closure_i \times PrePost_t + \delta_i + \mu_t + u \qquad \text{Eq. 2}$$

Where Y_{it} is the KDE of shootings for each school i, $Closure_i$ is a binary indicator for whether each school i was vacant or not, and $PrePost_t$ is a binary indicator of the post- and pre-intervention periods. The interaction term $\beta_3 Closure_i \times PrePost_t$ provides the DiD effect of closure on firearm violence. The term u is a residual term representing the spatial error term as $u = \rho Wu + \varepsilon_{it}$, where ρ is the spatial autoregressive parameter, W remains the spatial weights matrix, and ε_{it} is an additional residual. This model was estimated by maximum likelihood using the spml(.) function from the splm package (Millo and Piras, 2012).

2.4. Stability analysis: alternative balancing procedure

As a stability analysis, an alternative procedure was used to generate a balanced dataset and fulfill the stable unit treatment value assumption. In this analysis, we implemented a matching procedure utilizing a propensity score nearest matching algorithm with Mahalanobis distance calculation at a 3:1 ratio. To ensure the Stable Unit Treatment Values Assumption was satisfied, matched schools could not be selected from within 0.25 miles of a vacant school in the final dataset. This eliminated 7 of 405 elementary schools. Equations, balance statistics and results of this analysis are included in Appendix 1.

3. Results

3.1. Descriptive statistics

Census tracts of vacant buildings (73.33%) had a higher percentage of Black residents than census tracts of open schools (43.38%). These census tracts also had higher povery rates (Vacant - 33.89%, Open – 25.88%), Social Vulnerability Indices (Vacant - 0.81, Open - 0.70), with similar patterns observed for other variables. After weighting, balanced demographics were achieved in both samples, with each standard mean difference for all variables less than 0.1, a cut-off indicating appropriate balance in the literature (See Fig. 2 for balance plots demonstrating standard mean differences prior to and following weighting procedures) (see Fig. 3).

Mean kernel density estimates of shootings at the beginning and end of the period (2010-2019) are also shown in Appendix Table 1, demonstrating an increase in shootings around vacant school buildings. The difference between the mean shootings per square mile of vacant schools and open schools was 9.8 in 2010 prior to school closure and 13.31 in 2019 following school closure. In absolute terms, vacant buildings saw an increase in the shootings per square mile, while open schools saw a small decrease in shootings.

3.2. Firearm violence related outcomes

Consistent with the assumptions for DiD analyses, pre-intervention trends in firearm violence outcomes (assaults/batteries, shootings, weapons violations) were not significantly different between treated and matched non-treated group based on visual inspection (Fig. 1) and formal testing utilizing Eq. (1).

In the weighted DiD analysis (Table 1), initial building vacancy was associated with a 9.88% increase in shootings near the school location (Model 1 95% CI: 3.59-16.99; p = 0.001). Building vacancy was also associated with increased assaults/batteries with a firearm (5.46%; 95% CI: 1.14, 10.17; p = 0.012) and increased weapons violations (6.48%; 95% CI: 0.52, 13.18; p = 0.032).

The sample was then separated to explore different post-treatment dynamics. Long-term building vacancy was associated with a 10.20% increase in shootings (Model 2 95% CI: 2.73-18.84; p = 0.005), 5.76% increase in assaults/batteries with a firearm (Model 2 95% CI: 0.53, 11.56; p = 0.030), and an 8.21% increase in weapons violations (Model 2 95% CI: 1.08, 16.42; p = 0.022). Buildings that were re-used did not experience statistically significant increases in shootings (p = 0.06), assault/battery with a firearm (p = 0.14), or weapons violations (p = 0.53).

These results were found to be stable to an alternative balancing

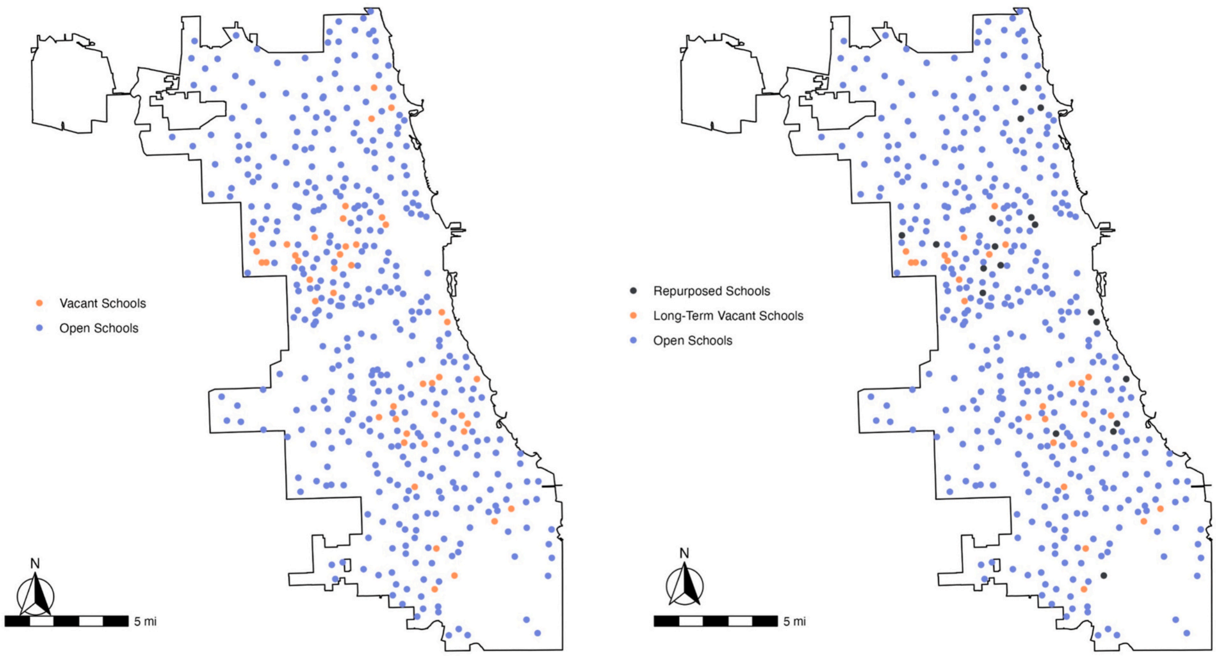

Fig. 1. Locations of vacant and control schools for analysis.

procedure utilizing matching and alternative modeling with a mixed-effects ordinary least squares difference-in-difference model (Appendix 1). Slightly larger estimates of the association between school building closure and firearm violence were produced with this alternative procedure. Additional sensitivity analyses were conducted on shootings to assess stability of results to changes in KDE bandwidth selection and time period. In Appendix Table 3 results were found to be stable to variation in the bandwidth utilized to generate kernel density estimates (0.25 miles to 2 miles). Increasing bandwidth tended to decrease the magnitude of the observed association but decrease the uncertainty in the estimate, likely owing to improved stability of estimates when smoothing individual events over larger areas. Results were also found to be consistent across alternative time periods in Appendix Table 4. Two alternate time periods, from 2010 to 2017 and from 2010 to 2023 were included, with estimates produced within a one percentage point difference from the primary analysis (2010-2019).

4. Discussion

In this propensity-score weighted difference-in-differences analysis of the 2013 mass public school closure in Chicago, closure was associated with significant increases in firearm violence in the neighborhoods surrounding vacant schools. This study is among the first to explicitly analyze firearm violence outcomes in the context of mass school closure over a 10-year study period. The relationship between school closures and increased firearm violence was present among vacant school buildings, but was not demonstrated in the analysis of school buildings that were re-used. We theorize that these disparate findings highlight the importance of productive re-use after school closure, and correspondingly, the potentially deleterious effects of building vacancy and disinvestment in affected communities.

Prior research on mass public school closure has described how public schools and their buildings are community spaces for social connection, supporting neighborhood organizations, and anchoring economic institutions (Ewing, 2018; Good, 2022). Beyond educational activities, school closure causes a loss of public space within which residents and students may come together for various activities (e.g. Parent Teacher Associations, sporting events, after-school activities, community meetings). Others have suggested that school closures can be viewed as a loss of community control (Brinig and Garnett, 2009; Rich and Owens, 2023), with physical manifestations including a loss of parking guards and safety personnel that monitor youth transitions to and from schools. It can also represent a symbolic loss for a community, with a school carrying the weight of community pride and a claim by a neighborhood to the future of a city (Ewing, 2018). Together, the loss of these community-level factors could plausibly reduce collective efficacy, social cohesion, and ultimately, social control. Social characteristics like collective efficacy have repeatedly been demonstrated to mediate relationships between neighborhood disadvantage and community violence (Roman, 2002). Importantly, these dynamics do not exist in a vacuum. In *Ghosts of the Schoolyard*, Ewing emphasizes the way in which school closure represents an additional disruption to community stability, undertaken primarily in racially minoritized communities that have already experienced multiple similar disruptions, such as public housing privatization and mass incarceration (Ewing, 2018).

Our findings also suggest that the physical abandonment of schools leading to long-term vacancy is potentially a key factor contributing to increased firearm violence. Past research has demonstrated the relationship between a disorganized, deteriorated physical environment and increased burden of firearm violence, with much of the literature focused on the impact of greening vacant lots and housing remediation (Branas et al., 2011; Moyer et al., 2019; South et al., 2023; Asa et al., 2026). In separate analyses, we observe a weaker association between buildings that were subsequently re-used and firearm violence. This suggests that physical vacancy is particularly important in the relationship between school closure and firearm violence, and productive re-use could minimize the observed finding.

Contrary to this study, some previous studies of mass school closure have found short-term decreases in the rates of overall crime and violent crime. Our findings indicate that firearm shootings, assault/battery with a firearm, and weapons violations increased over a prolonged period around vacant schools. One factor that may contribute to this difference is the spatial scale that was selected for analysis. Past studies of vacant schools have focused primarily on the physical disorder that arises from vacancy by assessing changes in crime within very small radii from the site of the closed school, generally on the same block or part of a block. In this analysis, we assessed vacancy across geographic areas by utilizing a kernel density estimate with a primary bandwidth of 0.5 miles (Good,

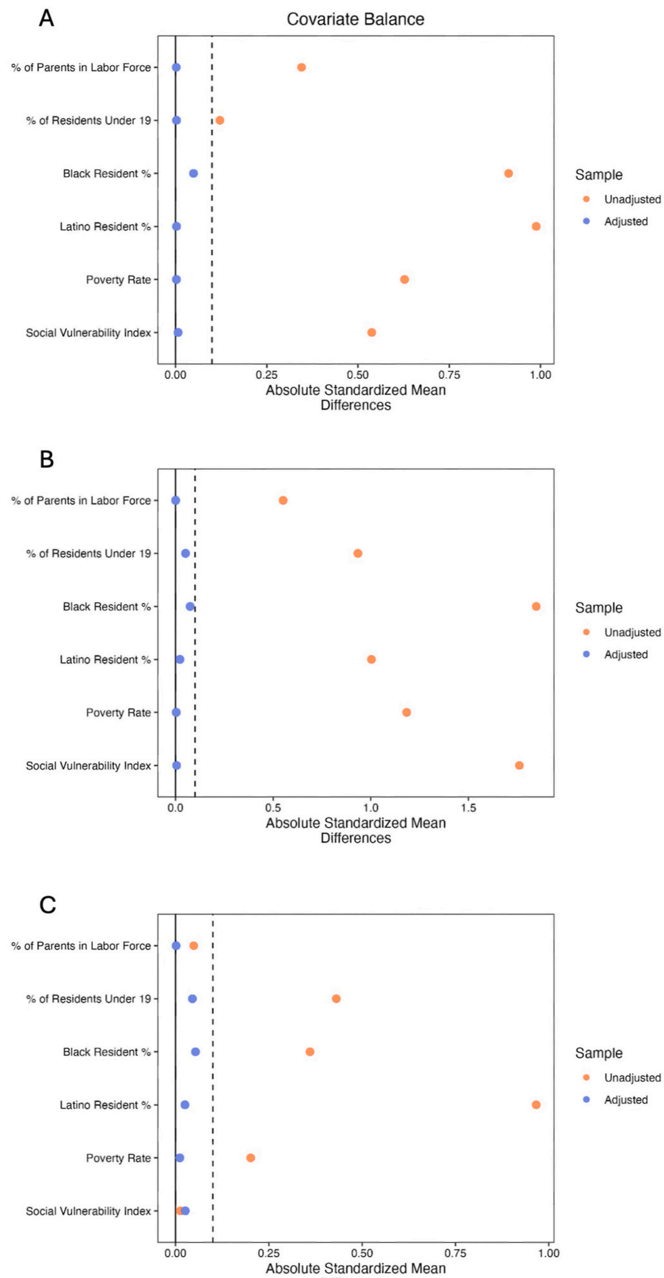

Fig. 2. Balance plots demonstrating pre- and post-balance of considered variables from the American Community Survey for A) Model 1: Vacant School Buildings, B) Model 2: Long-Term Vacant Schools, C) Model 3: Repurposed Schools.

2022; Grant et al., 2017; Pearman and Marie Greene, 2022; Rich and Owens, 2023). This allows for balance between capturing the neighborhood-level effects of school closure, while minimizing the potential for over-smoothing and loss of heterogeneity. We further demonstrate stability of results with varying bandwidths from 0.25 miles to 2 miles, with diminishing effect sizes at larger bandwidths, as would be expected with diffusion of the effect of school vacancy over a larger area.

Another key difference from existing literature is that we analyzed data over a longer period of time, including data up to 6 years after closure in the primary analysis. This time-scale was selected to allow for examining the long-term effects of increasing physical disrepair and theorized changes in neighborhood dynamics secondary to closure. Further, we have included sensitivity analyses demonstrating overall stability of the results over shorter and longer time periods. Firearm-related violence is of unique relevance to youth compared to other forms of crime or even violence (e.g., domestic violence). Youth are at particularly high risk of mortality via firearm shootings, which have now surpassed motor vehicle accidents as the single most frequent cause of death amongst U.S. children. Changes in neighborhood conditions and related social environment factors may effect youth, contributing to cycles of firearm violence in particular.

It is important to consider the degree to which these results can be interpreted as causal effects. Because closures and subsequent vacancies were determined through a district-wide administrative process and were implemented simultaneously across schools, the design provides quasi-experimental inference for assessing the impact of closure. The observed increases in firearm violence are consistent with a causal effect of school closure designation, conditional on the assumption that, after matching and adjustment, vacant and comparison schools would have followed similar trends in violence in the absence of closure. A strength of this analysis is that while it is difficult to exclude the potential impact of unobserved differences between treatment and comparison groups, the inverse probability weighting procedure (and, in stability analysis, the nearest-neighbor matching procedure) generated well balanced comparison groups. Variables chosen for weighting demonstrated standard mean differences no greater than the accepted standard of 0.10, with most being below 0.05.

However there are key limitations in our ability to interpret these results as causal. Schools were selected for closure based in part on factors such as performance and enrollment that may reflect underlying neighborhood dynamics (De la Torre et al., 2015), and we cannot rule out the possibility that unobserved, time-varying factors influenced both closure decisions and subsequent trends in violence. In addition, building vacancy does not correspond to a uniform exposure at the building level: some vacant buildings were eventually repurposed. Secondary analyses explored this heterogeneity by distinguishing between post-closure building trajectories, but this heterogeneity also limits the confidence with which causal conclusions can be made. Additionally, while we selected a longer time-scale in order to capture potentially slow-evolving social and environmental dynamics within neighborhoods, this also limits causal interpretation by increasing the potential for additional differential shocks. However, even given these inherent limitations in the appropriateness of identifying causal effects of school closure from this analysis, our results at minimum demonstrate significant association between spatial areas that experienced closure and increased firearm violence. This illustrates concomitant disinvestment through school closure in physical space, social structures, and public resources in communities that were likely at greatest need for investment given the heavy and increasing burden of violence they experienced in the years following closure.

This quasi-experimental analysis of mass public school closure demonstrates the importance of public investment in neighborhoods and the important protective roles public schools may play in mitigating firearm violence. Multiple U.S. cities are once again considering significant school closures. While the decision to close a school is extremely complex and politically-charged, this study indicates that there may be broad consequences—including potential increases in firearm violence—if schools are vacated without concomitant plans for community investment and development. Our results particularly emphasize this given the null findings for repurposed school buildings. School districts have often argued for closure based on underutilization in a given building and a proposed ability to provide improved educational access elsewhere. However, educational access for an individual student may not account for broader neighborhood- and community-level impacts that also meaningfully affect students' ability to engage in education (Lofton and Davis, 2015). If school closure is deemed necessary, then it should be undertaken carefully to prevent the multifactorial impacts of closure on a community including increased vacancy, loss of a symbolic marker of community, and loss of a space to support

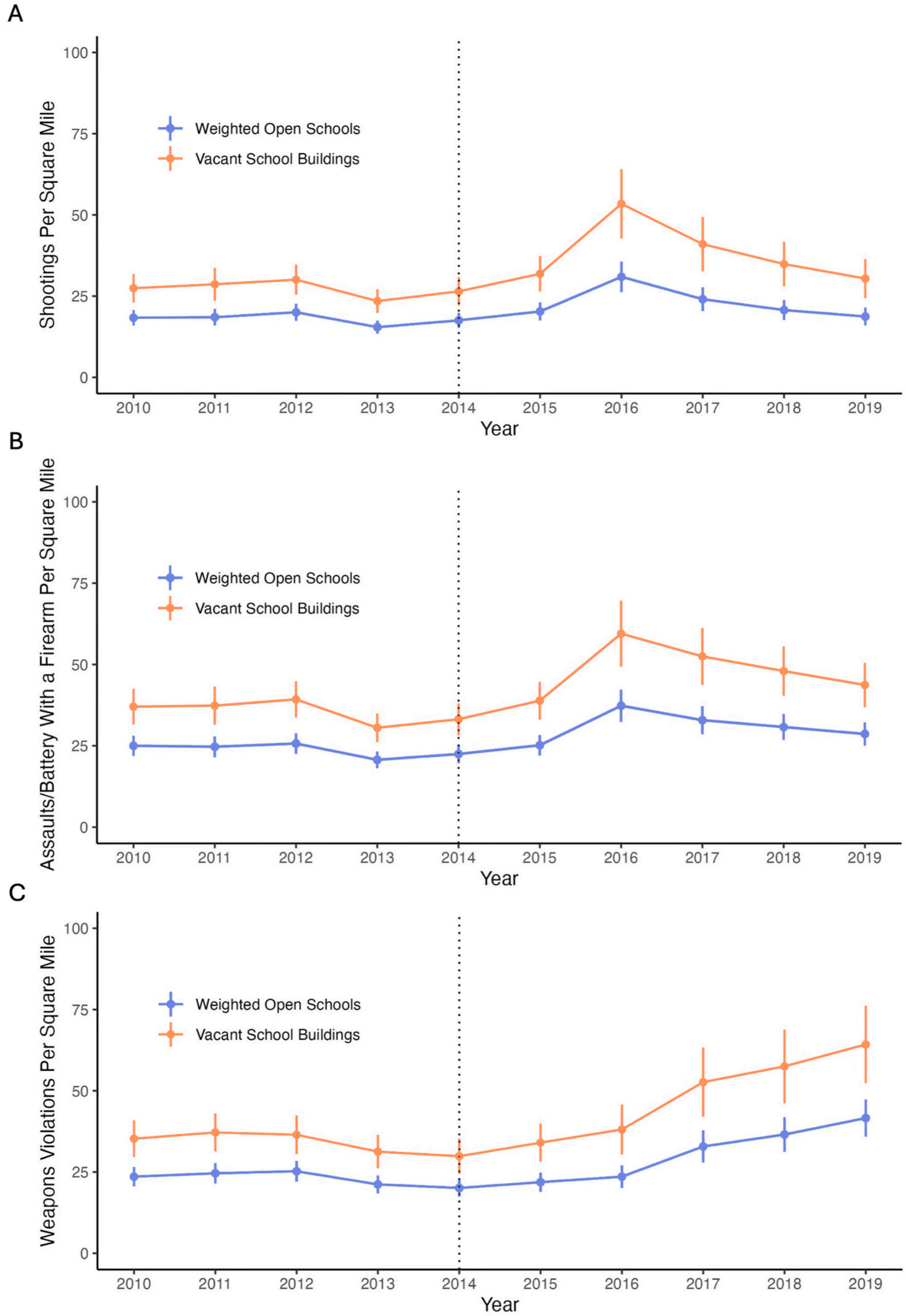

Fig. 3. Mean annual shooting density (shootings per square mile) around vacant and matched control schools over the study period in Chicago.

community building. Recent efforts in Baltimore offer an innovative structural approach to address the issue of long-term vacant school building closures. The *In All Ways Human* campaign is a multi-level, multimodal, youth violence prevention project that disrupts dehumanizing racial narratives of Black male criminality through visual storytelling and narrative change (Smith Lee et al., 2025). As part of the campaign, a large portrait mural was installed on the site of a long-term vacant elementary school. The mural contains portraits of Black boys and men, including graduates from the school spanning the 1950s-2000s. The project provided community members an opportunity to reclaim and transform the vacant school (a site of neighborhood disinvestment) to a space that affirms the idea that Black boys, men, and families are deserving of safety, economic stability, and social well-being. The *In All Ways Human* campaign is an emblematic example of a community-led project that challenges racialized spatial stigma; a neighborhood-level social process associated with firearm violence (Uzzi et al., 2024). To promote community safety, such an effort should be undertaken in conjunction with long-term reactivation and intentional investment in communities affected by school closures (Rajan et al., 2022).

5. Conclusion

In this quasi-experimental study of school closures in Chicago, we found that school closure and subsequent building vacancy were associated with significant increases in firearm violence in affected neighborhoods, and that this association appeared to be partially mitigated by subsequent school buildings re-use. Our findings highlight the important

Table 1
Percent change in density (kernel density estimate) of outcome variables in Difference-in-Difference analysis for vacant schools (Model 1) following closure compared to weighted non-vacant schools in Chicago. Models 2 and 3 delineate results by post-treatment status of long-term vacant or re-used, respectively.

Characteristic	Coefficient (95% CI)	% Change	p-value
Model 1: Vacant School Builidngs			
Shootings Per Square Mile	3.26 (1.26-5.27)	9.88 (3.59-16.99)	0.001
Assault/Battery	2.38 (0.52, 4.25)	5.46 (1.14, 10.17)	0.012
Weapons Violations	2.80 (0.24, 5.36)	6.48 (0.52, 13.18)	0.032
Models 2, 3: Categorical Treatment			
Model 2: Long-Term Vacant Schools			
Shootings Per Square Mile	3.99 (1.14-6.83)	10.20 (2.73-18.84)	0.005
Assault/Battery	2.94 (0.28, 5.59)	5.76 (0.53, 11.56)	0.030
Weapons Violations	4.25 (0.60, 7.91)	8.21 (1.08, 16.42)	0.022
Model 3: Repurposed Schools			
Shootings Per Square Mile	2.35 (−0.15, 4.87)	9.53 (−0.05, 21.91)	0.06
Assault/Battery	1.76 (−0.59, 4.12)	5.31 (−1.66, 13.35)	0.14
Weapons Violations	1.01 (−2.18, 4.20)	3.32 (−6.31, 14.91)	0.53

role that public schools may play in community safety, and we theorize that this association may be due to impacts on neighborhood social characteristics like collective efficacy and social cohesion, in addition to neighborhood blight and a lack of productive re-use of public space. Future research should seek to analyze the effect of school closure on neighborhood social characteristics, as well as explore the role of still-open schools in potentially mitigating the impacts of closure in nearby communities. Policymakers should carefully consider decisions around school closure. Moreover, policymakers, community residents, and community organizations should collaborate to co-create and co-implement strategies that mitigate potential harms of school closure and prioritize the safety and social well-being of students and the communities they live in.

Ethics declaration

Only publicly available, non-identifiable data was used in the study. For this reason, ethics approval through an institutional review board was not required.

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CRediT authorship contribution statement

Thomas Statchen: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization. **Michael Desjardins:** Formal analysis, Methodology. **Elizabeth Wagner:** Methodology, Writing – original draft, Writing – review & editing. **Odis Johnson:** Methodology, Writing – original draft, Writing – review & editing. **Elizabeth L. Tung:** Methodology, Writing – original draft, Writing – review & editing. **Mudia Uzzi:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.socscimed.2026.119428.

Data availability

The data is publicly available through the city of Chicago

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